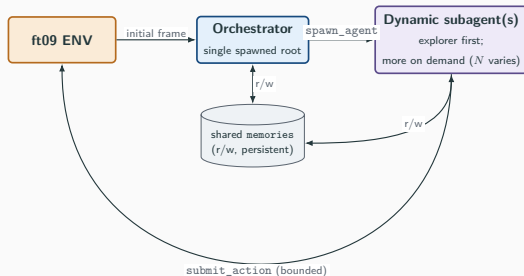


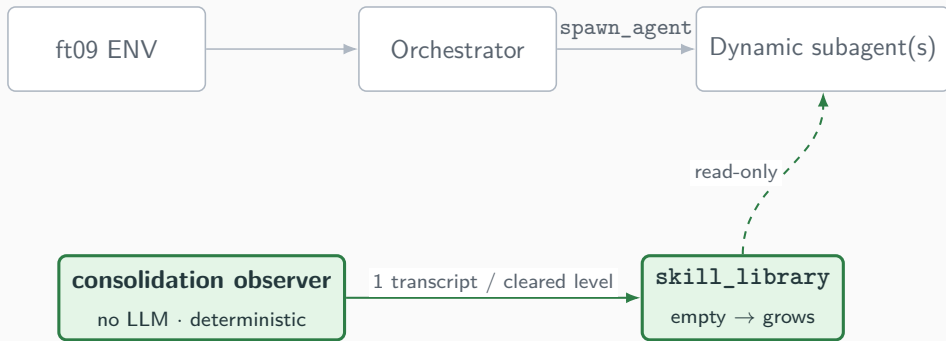
Original Symbolica: one orchestrator-rooted LLM tree

infra (unchanged): LLM calls $\rightarrow$ agentica-server:2345 $\rightarrow$ TRAPI proxy:9091 (Bearer $\rightarrow$ AAD) $\rightarrow$
agl-dev / TRAPI gpt-5.5



No skills: byte-identical original Symbolica — one orchestrator dynamically spawns subagents sharing memories, acting via bounded `submit_action`.

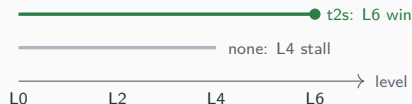
Trace2Skill: same loop + one additive skill directory



Trace2Skill adds only a fresh per-run **skill_library**: a deterministic observer records each cleared level's **verbatim action transcript**. Trace-verified: this text re-enters the LLM `body.input` after consolidation (87/145); causal reliance *not* tested.

Observed: $\text{trace2skill} \geq \text{none}$ on level (5/5, directional)

same-seed $\text{trace2skill} \geq \text{none}$ on level	5/5
s47 trace2skill	clean L6 WIN
s47 none	stalled L4
skill text in LLM body.input (post-consolidation)	trace-verified 87/145
causal reliance on skill	not tested
evidence strength	8/9 M3-censored
claim boundary	not clean-rev4



s47: an empty directory grew to **5** verbatim action transcripts that re-enter the LLM input;
 trace2skill reached **L6** where none stalled at **L4** — directional, causal use untested.